

Investigation of Robot Humour Styles on User Acceptance in a Gamified HRI Environment

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Abstract— Social humanoid robots should be equipped with natural communication tools, such as humour, to ensure seamless and intuitive interactions with human users. However, there is a limited understanding of how contrasting humour styles influence human decision-making, and current systems lack integrated, non-contact tracking frameworks to evaluate true user acceptance. Therefore, this project evaluates the impact of a humanoid robot's optimistic and pessimistic humour styles on human behavioral compliance and subjective user experience. A custom gamified Pygame simulation was utilized alongside a physical UBTECH Alpha Mini robot to deliver rule-based persuasive interventions. Additionally, a concurrent, non-contact computer vision pipeline utilizing OpenCV and DeepFace was implemented to estimate the user's real-time arousal and facial valence without requiring wearable sensors. The framework was tested on 30 participants in a controlled, between-subjects experimental design. The results demonstrated that the pessimistic humour style generated a moderate increase in behavioral compliance, but this yielding was driven by emotional detachment and a significantly higher frequency of neutral facial states. Conversely, the optimistic humour style encouraged users to resist persuasion and retain their original choices, while fostering a significantly happier sustained emotional state and yielding notably higher post-game user experience ratings.

Keywords— *Human-Robot Interaction, Persuasive Robotics, Humour Styles, Affective Computing, Computer Vision, Behavioral Compliance*

I. INTRODUCTION

In the context of Human-Robot Interaction (HRI), integrating human social tools into system programming has been commonly practiced to facilitate natural communication and influence human decisions for many years. Persuasive robotics and dialogue management are umbrella terms that are used for developing not only basic task execution but also fostering social connection and subjective user acceptance through specific conversational strategies like humour. The social robotics context in the modern collaborative environment is more than executing technical commands alone. Moreover, beside functional efficiency, humanoid systems are required to be competent in persuasive communication, which is essential for building trust, providing guidance, and seamlessly cooperating with human users in real-world settings. Therefore, engineers are constantly upgrading conversational algorithms and non-contact physiological tracking frameworks to enhance the robot's persuasive capabilities and accurately evaluate how human users cope with different social cues.

This study aims to propose a unified mechatronic framework that has been implemented using a gamified

Pygame simulation and a physical UBTECH Alpha Mini robot. The main objective of the paper is to ascertain the effectiveness of a rule-based conversational model which entirely focuses on optimistic and pessimistic humour styles in influencing students' behavioral compliance, emotional arousal, and overall user acceptance. The presentation of the paper is as follows. First, it outlines the theoretical overview providing the rationale for the conduct of persuasive robotics and vision-based physiological tracking in HRI environments. Second, it elaborates on the methodological tenets employed to collect and analyze data in this study. In that section, the proposed digital narrative environment and the concurrent OpenCV computer vision pipeline that were implemented are presented. Then, the quantitative findings of the study are presented in the subsequent section. Afterwards, the findings are discussed in light of relevant objective choice shifts and subjective emotional states. Some concluding remarks are provided at the end of the paper.

II. LITERATURE REVIEW

According to Bartneck et al. [1], Human-Robot Interaction (HRI) is built around core principles, one of which states that engineers must combine computer science and psychology to design robots that fit easily into human lives. It was cited by Fong et al. [2] that social robots are a foundational model of this interaction. The terms social robotics and persuasive robotics are interchangeably used in this paper to describe machines that intentionally change a user's choices, beliefs, or behaviors without forcing them [3].

There are primary technical mechanisms of dialogue management in HRI suggested by Traum and Larsson [4]: tracking conversational context, interpreting user input, choosing correct timing, and coordinating physical responses. Out of these mechanisms, the focus is heavily related to developing rule-based algorithms for persuasive responses. These mechanisms are utilized to prepare robots for real-world interactions and offer them better capabilities in guiding human decisions. Through the use of natural spoken conversation, it enables humanoid systems to develop communication skills which emphasize effective persuasion, motivating users to alter their choices by building trust and engagement quickly [5], [6].

With reference to Martin et al. [7] and Russell [8], the psychological frameworks of humour and affect are focused on distinct models: the Humour Styles Questionnaire (HSQ) and the Circumplex Model of Affect. Regarding HRI and emotional tracking goals, there is an emphasis on estimating physiological arousal via Eye Blink Rate (EBR) and facial valence using computer vision [9], while persuasive robotics

emphasizes the acquisition of behavioral compliance. In addition to these main objectives of HRI tracking, according to Venâs et al. [10] and Faber et al. [11], testing users in a structured decision-rule simulation can also be called narrative gamification, which mitigates performance anxiety.

Regarding the categorization of robot humour, both positive and negative communication styles are emphasized [12]. There are four fundamental dimensions of humour in the HSQ: affiliative, self-enhancing, aggressive, and self-defeating. Out of these dimensions, a condensed binary model of optimistic (derived from self-enhancing) and pessimistic (derived from self-defeating) humour was adopted in this study, which focuses on evaluating how linguistic framing influences user compliance. Gamified "food selection" decision tasks were used to help track the students' immediate choice shifts, emotional arousal, and facial valence using non-contact computer vision pipelines without the need for wearable sensors.:

- 1) **Hypothesis 1 (H1):** An optimistic robotic humour style will result in higher subjective user experience and satisfaction scores.
- 2) **Hypothesis 2 (H2):** A pessimistic robotic humour style will trigger a higher rate of behavioral compliance, resulting in more frequent choice shifts.

III. METHODOLOGY

An experimental study was conducted for 30 Mechatronics Engineering students at the International Islamic University Malaysia (IIUM). It was a highly controlled laboratory session in the 2025-2026 academic year. The experimental protocol constitutes a 15 to 20-minute session per participant. The framework and testing variables used are exclusively related to "Persuasive Robotics".

To find out participants' perceptions of the proposed robotic intervention and system design, users completed a 24-item, 5-point Likert scale questionnaire. The questionnaire used in the study was integrated directly into the Pygame interface. Participants were instructed to rate the statements as 5 for strongly agree to 1 for strongly disagree, comparing their User Interface clarity, In-Game Experience, and Post-Game Perception.

A rule-based conversational model utilizing a between-subjects experimental design was adopted in this study. In the assessment pipeline, there are concurrent computer vision metrics to find out what types of physiological states users experience in the Human-Robot Interaction (HRI) environment and how they felt during the decision-making nodes.

A. The Proposed System Architecture

Components that were used in the mechatronic framework are related to 'Human-Robot Interaction'. Some of which are extracted from established tracking libraries: OpenCV and DeepFace. The main components included in the proposed system are described in the following.

- 1) **Gamified Narrative Simulation:**
Under this component, participants were engaged with general spheres of digital navigation, food-preparation decision nodes, and narrative interactions using a Pygame simulation titled Foodie Adventure.



Fig.1 Core narrative environment

- 2) **Humanoid Robot Intervention:**
Regarding persuasive communication skills, users were made aware of a physical UBTECH Alpha Mini humanoid robot based on a rule-based conversational model. The robot was actuated via a WebSocket bridge to deliver strict optimistic or pessimistic scripts.



Fig.2 UBTECH Alpha Mini Robot

- 3) **Vision-Based Affect Tracking:**
For physiological tracking, a three-tier machine learning hierarchy, bounding box localization, Haar Cascades, and DeepFace neural networks, and their strengths in real-time execution are included. Under this component, the system calculates physiological arousal using normalized Eye Blink Rates (EBR) and average blink duration as shown in the following equation:

$$Arousal = 0.7 \times br_{norm} + 0.3 \times bd_{norm} - 0.15$$

where br_{norm} is the normalized blink rate, bd_{norm} is the normalized average blink duration, and the output is bounded to a real-time approximation of emotional intensity.

B. Experimental Procedures

A strictly controlled chronological procedure was extensively used. For each decision task, the system managed the interaction giving more priority to the robot's auditory intervention than user input time. Users were engaged with a "Brain Lock" protocol for pair-wise choice tasks and persuasion sessions after every initial selection. Tutorial scenarios, informed consent briefings, and standardized narrative loops were shared with the participants prior to the main testing time.

C. Assesment System

- 1) **Behavioral Compliance Tracking:**

Decision shifts were assigned as objective boolean variables within the central Python execution script. Participants are required to interact with the 10 food vendors to be able to register their initial and final selections, calculating a definitive compliance ratio.

- 2) Automated Metric Aggregation for Affect: Concurrent computer vision pipelines are arranged to test participants' emotional arousal, valence, and sustained affective states. The system also has to deal with fusing these metrics into a single categorical state (e.g., Happy, Neutral, Fear) mapped to the Circumplex Model of Affect.

In a persuasive robotics focused HRI system, there may be varieties of communication strategies to be used. However, the proposed framework focused on 'Optimistic' and 'Pessimistic' related humour styles only. The reason of choosing these two contrasting styles particularly is because it was intended that researchers will become more aware of how positive reinforcement versus risk-focused framing influences human behavioral compliance and subjective social judgment.

IV. FINDINGS

According to the findings of the system evaluation and reliability analysis, the proposed mechatronic framework successfully rendered the 2D narrative environment without processing latency. All auditory exposition and storyline narration were successfully delegated directly to the physical Alpha Mini robot via a WebSocket bridge. To ensure the internal consistency of the questionnaires used during this experimental phase, TABLE I describes the Cronbach's Alpha scores across three categories: User Interface (UI), In-Game Experience, and Post-Game Perception.

TABLE I. Cronbach's Alpha Reliability Scores

Questionnaire	Cronbach's Alpha	Reliability Level
User Interface (UI)	0.664	Acceptable
In-Game	0.610	Acceptable
Post-Game	0.787	Good

It was found that the survey instruments were suitably reliable for evaluating the user experience. In this proposed experimental design, normal data distribution was confirmed ($p > 0.05$) for parametric testing, while a non-parametric Mann-Whitney U test was applied for evaluating the frequency counts of sustained emotional states.

A. Behavioral Compliance and Emotional States

To investigate participants' overall behavioral compliance in response to the robotic intervention, decision changes and decisions kept are analyzed. TABLE II describes the independent-samples t-test results comparing the Optimistic and Pessimistic test groups.

TABLE II Independent-Samples T-Test for Behavioral Compliance

Metric	Optimistic (Mean/SD)	Pessimistic (Mean/SD)	t-value	p-value	Effect Size (d)
Total Decision Changes	3.47 / 2.13	4.80 / 2.48	-1.577	0.126	0.58
Total Decisions Kept	6.53 / 2.13	5.20 / 2.48	1.577	0.126	0.58

It was found that while the p-value ($p = 0.126$) did not reach the strict statistical significance threshold, the robot's humour style exhibited a moderate practical significance ($d = 0.58$) on user decision-making behavior. Specifically, 65.3% of the measured retention instances ($M = 6.53$) showed that participants interacting with the optimistic robot were more likely to resist persuasion and keep their original choices. Conversely, users interacting with the pessimistic robot trended toward yielding to the robot and changing their decisions more frequently ($M = 4.80$).

The following figure describes the visual overlay capturing real-time physiological metrics, which supports this behavioral trend.

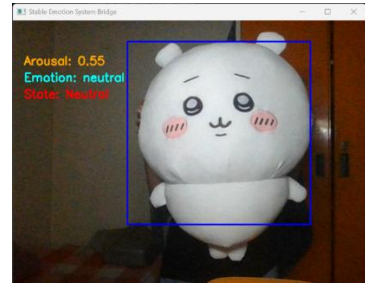


Fig.3 Real-time facial and EBR tracking

To contextualize why the pessimistic robot triggered more choice shifts, the concurrent computer vision data was analyzed. It was found that the pessimistic robot elicited 'Neutral' states significantly more often ($p = 0.033$) than the optimistic robot. There were no statistically significant differences found for negative emotions (Fear, Angry, Sad). This indicates that the higher rate of decision changes in the pessimistic condition was driven by a state of emotional detachment and risk aversion rather than active distress.

B. Subjective User Experience

Subjective user experience findings also support the behavioral data. Students' responses to the integrated Pygame questionnaire describe how the distinct humour styles influenced their overall perception of the interaction.

TABLE III Independent-Samples T-Test for User Experience

Metric	Optimistic (Mean/SD)	Pessimistic (Mean/SD)	t-value	p-value	Effect Size (d)
UI Rating	4.50 / 0.41	4.50 / 0.44	0.000	1.000	0.00
In-Game Rating	4.26 / 0.48	4.21 / 0.44	0.297	0.768	0.11
Post-Game Rating	4.05 / 0.58	3.58 / 0.75	1.909	0.067	0.70

Regarding participants' perceptions of the gamified environment, both groups rated the User Interface and in-

game mechanics identically. This identical baseline rating ($M = 4.50$) proves that the Pygame environment provided a consistent functional experience that did not bias the users' interaction across test conditions.

It was also discovered that the optimistic humour style successfully sustained positive user affect throughout the interaction. The analysis revealed a moderate-to-large effect size ($d = 0.70$) in post-game evaluation, approaching statistical significance ($p = 0.067$). This positive trend in post-game evaluation is strongly supported by the physiological data, which confirmed a statistically significant difference ($p = 0.037$) in the frequency of 'Happy' states for the optimistic group. Ultimately, the results demonstrate that while a pessimistic humour style is effective at forcing immediate task compliance, an optimistic humour style demonstrates a much stronger practical effect on long-term user acceptance and positive social judgment outcomes.

V. DISCUSSION

To test if an optimistic robot humour style leads to a better user experience and higher satisfaction, the quantitative and visual data were analyzed. The results support this prediction. Post-game survey scores showed strong positive responses to the optimistic robot. Computer vision data also showed that participants felt happy significantly more often during these interactions, validating the positive valence axis of the Circumplex Model of Affect [8]. However, User Interface ratings remained identical across both groups, proving that the high satisfaction came directly from the robot's social cues rather than the game design.

The behavioral tracking data helped evaluate the second hypothesis regarding the pessimistic robot. It was found that the pessimistic style caused higher behavioral compliance, making users change their decisions more frequently, which aligns with the compliance tracking metrics explored by Faber et al. [11]. Ocular and facial tracking data showed a high frequency of neutral expressions during these interactions. This suggests that users changed their minds due to risk aversion and emotional detachment rather than feeling active fear or distress, separating these reactions from classic negative affect frameworks [8].

Looking at how optimistic framing influenced choices, the data showed that most participants confidently kept their original decisions. They resisted persuasion because the optimistic robot provided reassuring support and empathy. Because participants were navigating a collaborative simulation, they felt like they were in a real partnership, matching the interactive field findings of Edirisinghe et al. [6]. The optimistic interaction made them feel more secure and reduced their psychological resistance to the robot's suggestions.

Out of various communication methods in persuasive robotics, specific humour styles are uniquely suited for different mechatronic goals, confirming that this study successfully achieved its three primary research objectives. Developing the Pygame simulation and integrating the physical Alpha Mini robot fulfilled the first two technical objectives by providing a highly reliable testing environment. The final objective was met through statistical analysis, proving that mapping emotions via the Circumplex Model of Affect [8] bridges the gap between user feelings and behavioral choice shifts. Therefore, this non-contact vision framework and rule-based conversational loop should be tested in other interactive environments, building upon the

baseline frameworks established by Martin et al. [7] and Edirisinghe et al. [6].

Finally, the different user reactions to optimistic and pessimistic persuasion reflect the core goals of modern HRI frameworks. This highlights why engineers must consider human psychological needs when designing robot dialogue systems, as discussed by Traum and Larsson [4]. Future mechatronics designers should focus on nuanced conversational algorithms to integrate social robots into daily life safely [1]. While a pessimistic style forces immediate task compliance, an optimistic humour style is essential for long-term user trust, genuine acceptance, and positive social judgment.

VI. CONCLUSION

The study undertaken contributes the positive outcomes of structured robotic humour styles in developing behavioral compliance and subjective user acceptance of Mechatronics Engineering students in the International Islamic University Malaysia. It was intended to propose a unified mechatronic framework that aims to develop not only adaptive task execution but also real-time physiological tracking in an interactive HRI environment. As the 21st century collaborative workplace demands social intelligence and adaptive communication skills beyond technical skills, dialogue management algorithms need to be upgraded according to the needs of the users. In this study, the majority of students perceived that the optimistic humour style is more engaging than traditional non-social robotic interactions and they found the system components used in the framework are interesting and helpful to improve their overall interactive experience. As a further study, users should be engaged with a long-term, multi-session interaction comparing pre-assigned robot profiles and user-selected personas as experimental research.

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